

Japan, used by the Europeans to pay for Chinese goods such as silk, lacquerware, and porcelain, stimulated economic growth, it also caused inflation.

Technological change

Ming China had inherited a legacy of scientific and technological innovation from the Song dynasty, which had left the country at the forefront of many scientific fields, including navigation and the military applications of gunpowder—a substance discovered during the Tang era whose use had spread to Europe from China in the 13th century. Under the Ming, though, the pace of progress slowed and by the later part of the dynasty, ideas had begun to flow in from Europe.

The Chinese military began to use artillery of European manufacture, and knowledge of European mathematics and astronomy was introduced to the country through Jesuit missionaries, including Matteo Ricci, who lived in Beijing from 1601 to 1610. He translated the ancient Greek mathematician Euclid's *Geometry* into Chinese, as well as a treatise on the astrolabe (an astronomical instrument used for taking the altitude of the sun or stars). In 1626, the German Jesuit Johann Adam Schall von Bell wrote the first treatise in Chinese on the telescope, bringing Heliocentrism (an astronomical model in which the sun lies at the center of the universe) to a Chinese audience.



On taking the throne, Hongwu issued his own traditional bronze coinage, although a shortage of metal led to the reinstatement of paper money, made of mulberry bark.

The Ming collapse

The late Ming began to suffer many of the same issues that had led to the fall of the Yuan. Crop failures reduced the productivity of China's vast agriculture, and famines and floods led to widespread unrest in rural areas. The army's pay began to fall into arrears, leading to discipline problems and desertions, while localized peasant uprisings coalesced into more general revolts. Meanwhile, on the northeastern frontier, the Manchus had built a state along Chinese lines at Mukden in Manchuria—calling their regime the Qing dynasty in 1636—and were now poised to take advantage of the Ming's imminent collapse. They were aided in this by a revolt led by Li Zicheng, a rebel leader whose forces entered Beijing in 1644 unopposed, prompting the emperor to commit suicide. In desperation, the Ming military called on the Manchus for help. The tribesmen swept into the capital and drove out the rebels, but then seized the throne, and proclaimed the Qing dynasty in China.

An enduring legacy

Although the Ming had fallen victim to an agrarian crisis that coincided with renewed nomadic activity on its frontiers, this was a combination that had also brought down dynasties before it. The bureaucracy that had given China centuries of constancy and reduced the possibility, or even the need, for internal dissent, was slow to adapt itself to times of fast-moving crisis.

Yet even so, the Ming era had brought great wealth and success to China. The population expanded from around 60 million at the start of its rule, to around three times that number by 1600. Much of this growth was centered in medium-sized market towns, rather than in large cities, and an increase in agricultural production led to the rise of an affluent merchant class in the provinces. Many of the elements of orderly government that Hongwu had inaugurated were carried over into the succeeding Qing dynasty, providing China with a degree of unity, stability, and prosperity that the European states of that period could only envy and admire. ■